



CSE 3313: Computer Architecture

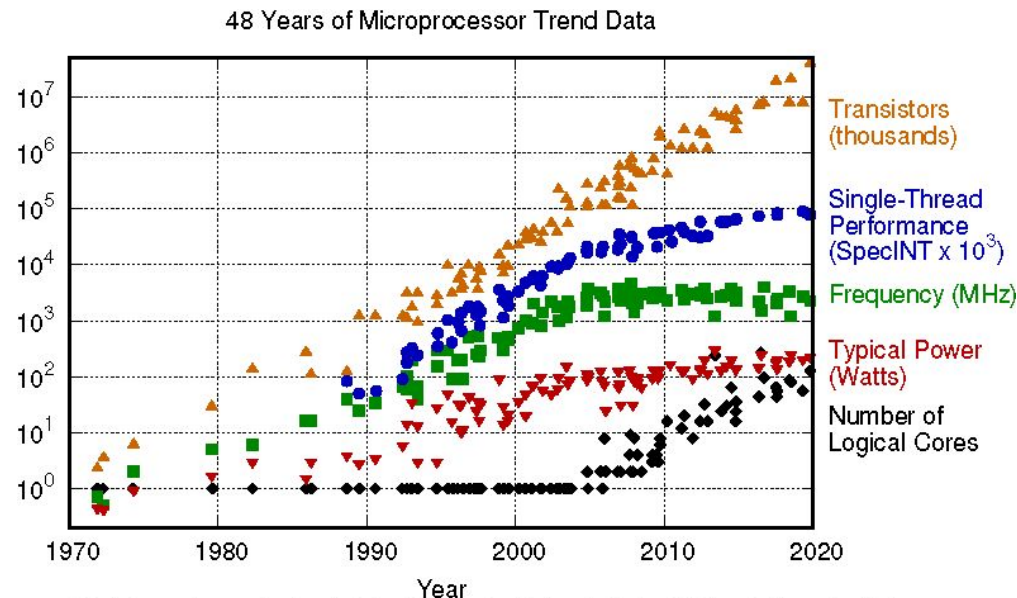
Sidratul Tanzila Tasmi

Lecturer, Department of Computer Science and Engineering
United International University

Innovative Drive for Computer Architecture



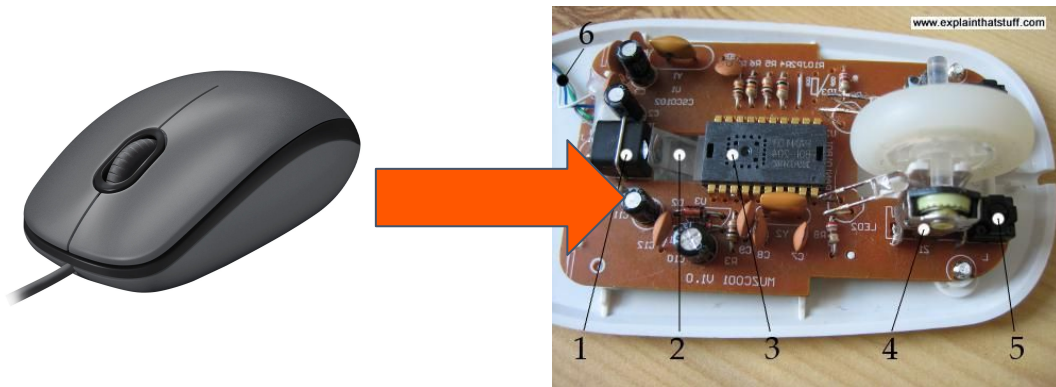
The number of transistors on a microchip doubles every year, while the cost is halved over the time frame.



Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten
New plot and data collected for 2010-2019 by K. Rupp

Innovative Drive for Computer Architecture

Using Abstraction for design simplification.



Innovative Drive for Computer Architecture

Performance through parallelism and pipelining

Single worker
digging the ground



**2 days to
complete**

More than 1
worker digging the
ground

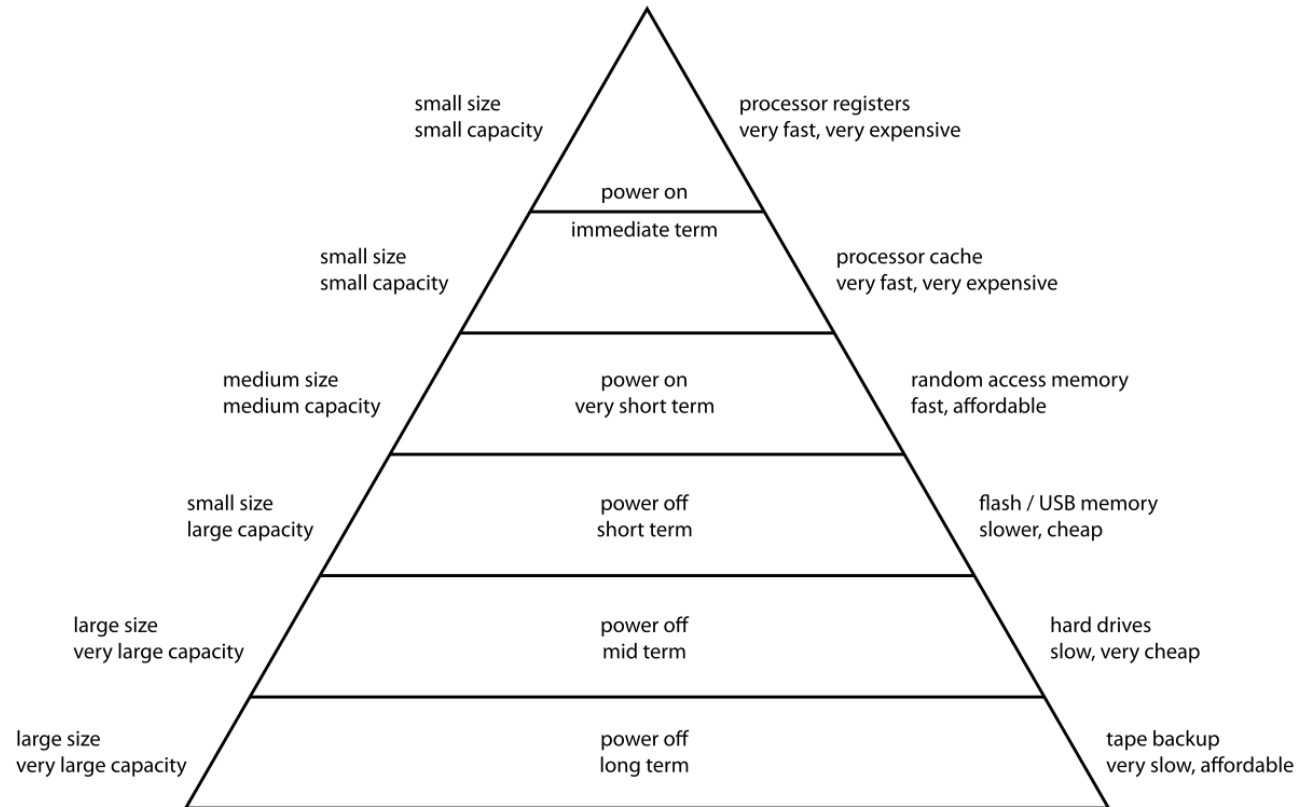


**1 day to
complete**

Innovative Drive for Computer Architecture



Computer Memory organization



After Seeing CA.

Syllabus

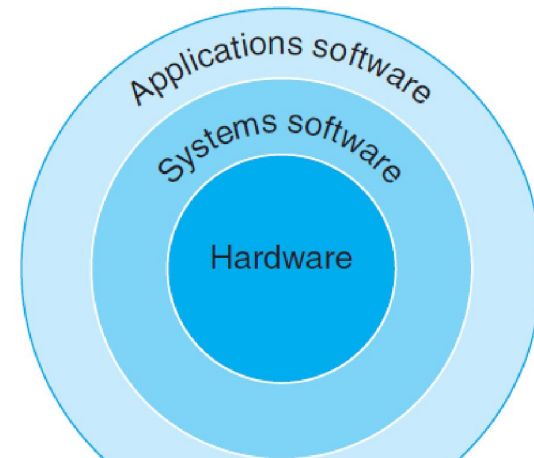


Arey mujhe chakkar aane laga hai

Software



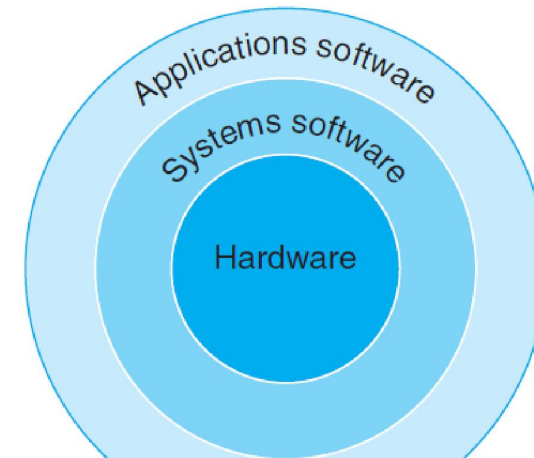
- Application software –
 - Word Processors, Email, Internet Browsers, Games
- Systems software–
 - Compilers, Operating Systems, device drivers



Hardware



- CPU (Central Processing Unit)
- Memory (RAM, ROM, Storage)
- I/O devices (mouse, keyboard, Monitor)



Basic Parts of a Computer

Central Processing Unit (CPU) : The CPU is the brain of the computer. All the important decisions and manipulations are carried out by the CPU. CPU is further divided into the Arithmetic Logic Unit(ALU) and the Control Unit(CU).

Motherboard: The motherboard is the main circuit that connects all the components of a computer.

Memory: The RAM(Random Access Memory) is used for quick access of data.

Basic Parts of a Computer

Storage: The storage components include Hard Disk Drives(HDD) and the Solid State drives(SDD). They are used for storing data for the long term.

Input devices: The various devices via which we can input data into the computer from the user are called input devices. Examples of input devices include keyboard, mouse, scanners.

Output devices: Output devices are the devices via which the computer gives out the output to the users. Examples of output devices include printers and speakers.

Operating System (OS)



- **Definition:**

- Interfaces between a user's program and the hardware and provides a variety of services and supervisory functions.

- **Functions:**

- handling basic input and output operations.
- allocating storage and memory.
- providing for sharing the computer among multiple applications using it simultaneously.

- Examples of operating systems: **Windows, Linux, and MacOS**

Compiler



- The translation of a program written in a high-level language, such as C or Java, into instructions that the hardware can execute.
- The translation from a high-level language program to hardware instructions is complex.

Microprocessor without Interlocked Pipeline Stages (MIPS)

High-level
language
program
(in C)

```
swap(int v[], int k)
{int temp;
  temp = v[k];
  v[k] = v[k+1];
  v[k+1] = temp;
}
```

Compiler

Assembly
language
program
(for MIPS)

```
swap:
  muli   $2, $5, 4
  add    $2, $4, $2
  lw     $15, 0($2)
  lw     $16, 4($2)
  sw     $16, 0($2)
  sw     $15, 4($2)
  jr     $31
```

Assembler

Binary machine
language
program
(for MIPS)

```
000000001010000100000000000011000
00000000000110000001100000100001
10001100011000100000000000000000
100011001111001000000000000000100
101011001111001000000000000000000
```

Don't Worry!
We will not do this!

Writing a program

Using a visual
programming
language

Copy code from
Stackoverflow

Writing the
program in C

Writing it in
x86 assembly

Writing assembly
directly in hex

Using a magnet to set
individual bits on a hard
drive to write the binary
of the program



- Assembler
 - A program that translates a symbolic version of instructions into the binary version.
- Assembly Language
 - A symbolic representation of machine instructions.



Thank you!